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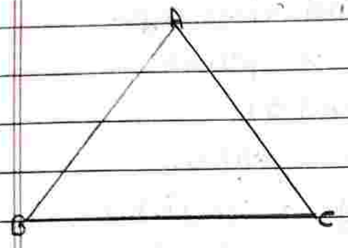
13. CONGUENCE OF TRIANGLE

★ Properties of Triangle

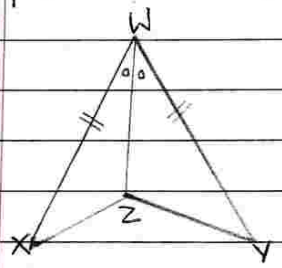


① Vertices - A, B, C
② Angle - ∠ABC, ∠A, ∠B, ∠C
③ Side - side AB, side AC, Side BC

Practice Set 13.1

1. In each pair of triangles in the following figures, part bearing identical marks are congruent. State the test and correspondence of vertices by which triangles in each pair are congruent.

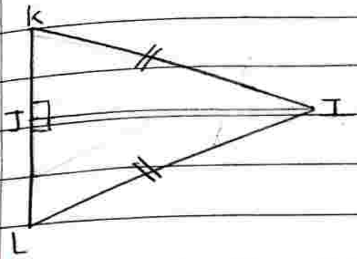
①



→ In, △XWZ & △YWZ

(i) ∠XWZ ≅ ∠YWZ ~~(Gig~~ (Given)
(ii) Seg WX ≅ Seg WY (Given)
(iii) Seg WZ ≅ Seg WZ (common side)

∴ By S-A-S test
△XWZ ≅ △XYZ
WXZ ⟷ WYZ

② 

→ In, $\triangle KIJ$ & $\triangle LIJ$

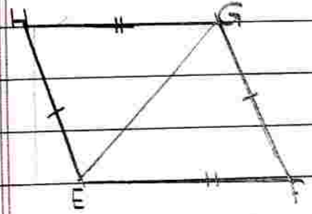
(i) $\angle IJK \cong \angle IJL$

(ii) seg IK $\cong$ $\angle IL$

By, hypotenuse - side test

$\triangle KIJ \cong \triangle LIJ$

$IKJ \longleftrightarrow ILJ$

③ 

→ In, $\triangle HEG$ & $\triangle FEG$

(i) seg HG $\cong$ seg EF

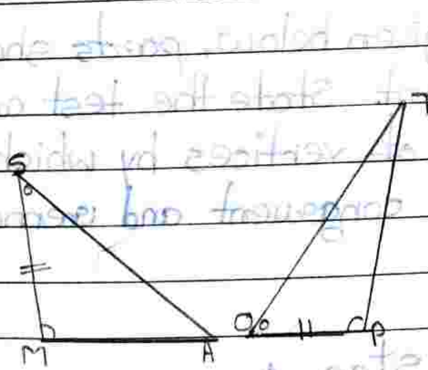
(ii) seg HE $\cong$ seg GF

(iii) seg GE $\cong$ seg GE

∴ By, S-S-S test

$\triangle HEG \cong \triangle FEG$

$GHE \longleftrightarrow GFE$

④ 

→ In, $\triangle SMA$ & $\triangle TOP$

(i) ~~seg~~ $\angle S \cong \angle O$

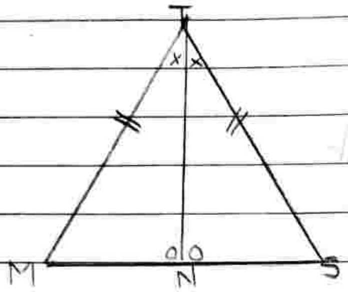
(ii) seg $SM \cong$ seg OP

(iii) $\angle M \cong \angle P$

$\therefore$ By, ASA Test

$\triangle SMA \cong \triangle TOP$

(5)



$\rightarrow$ In, $\triangle TMN$ & $\triangle TSN$

(i) seg $TM \cong$ seg TS

(ii) $\angle T \cong \angle T$

(iii) $\angle N \cong \angle N$

$\therefore$ By, S-A-A test

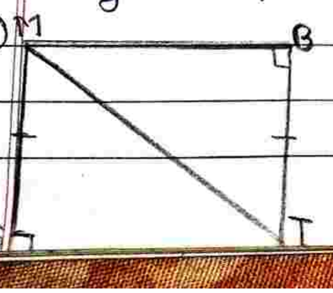
$\triangle TMN \cong \triangle TSN$

$TMN \longleftrightarrow TSN$

Practice Set 13.2

1. In each pair of triangles given below, parts shown by identical marks are congruent. State the test and the one to one correspondence of vertices by which triangles in each pair are congruent and remaining congruent parts.

(i)



Step 1

$\triangle MST$ and $\triangle MBT$

(i) Hypotenus $MT \cong$ Hypotenus MT

① Side MS ≅ Side BT

By

Hypotenus - Side Test

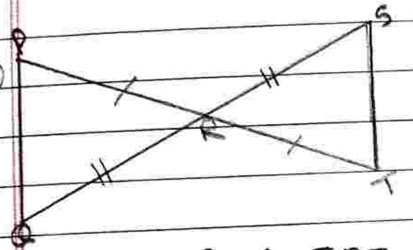
$\triangle MST \cong \triangle MBT$

$MST \longleftrightarrow MBT$

Corresponding Congruent Parts

Side ST ≅ Side MB

②



→ $\triangle PRQ$ & $\triangle SRT$

① seg PR ≅ seg RT

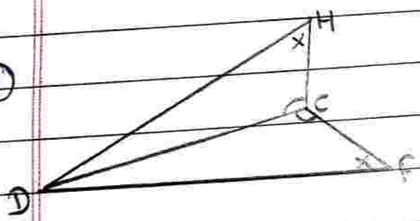
② $\angle PQR \cong \angle SRT$

③ seg QR ≅ seg RS

By, S-A-S test

$PRQ \longleftrightarrow SRT$

③



→ ~~△DHC & △F~~

$\triangle HDC$ & $\triangle FDC$

① $\angle H \cong \angle F$

② $\angle C \cong \angle C$

③ seg DC ≅ seg DC

By, A-S-A test

$HDC \longleftrightarrow FDC$

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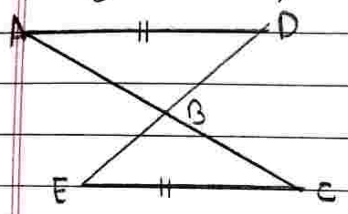
2* In the adjacent figure, seg AD ≅ seg EC Which additional information is needed to show that Δ ABD and ΔEBC will be congruent by A-A-S test ?



→ Δ ABD & EBC

(i) seg AD ≅ seg EC

(ii) ∠ ADB ≅ ∠ ECB

(iii) DAB ≅ CEB

By, A-A-S Test

ABD ⟷ EBC